

McDonald's



Social Media Command Centre

Checkpoint #4 — Final Evaluation & Findings

TEAM

Pranav C	Backend & AI Integration
Jeet T	Frontend & UI/UX
Ashish G	Data Engineering & Visualizations

IS492 · Spring 2026

MSIM'26 UIUC

 LIVE DEMO

`mcd-command-centre`
`.streamlit.app`

 GITHUB

`IS492-SP26/team-project-`
`social-media-command-centre`



Recap & Final Goals

CHECKPOINT 4

Problem Statement

32,736 customer reviews go largely unread by McDonald's branch management, and no existing tool transforms that feedback into direct operational action — current analytics flatten rich review signals into a single aggregate star rating that misleads decision-makers.

🎯 TARGET USERS

- McDonald's Branch Managers
- Regional Operations Leads
- Analytics-minded staff who review weekly performance

★ MOST REWARDING FINDING

"If I were a branch manager, I'd run this every Monday before my team huddle."

— P1, MSIM Student

4/5 users said they'd use the staff briefing memo in a real ops role

Evaluation Design Overview

METHOD

PARTICIPANTS

- N = 5 participants
- In-person think-aloud sessions
- ~25 minutes each
- MSIM, Business Analytics, Data Science, MBA (Ops)

STUDY TASKS (7 TASKS)

- T1: Find most-reviewed city
- T2: Identify worst branch
- T3: City filter interaction
- T4: Navigate to Co-Pilot
- T5: Generate AI insights
- T6: Download staff briefing
- T7: Adjust time range & regenerate

QUANTITATIVE METRICS

- Task success / assisted / failed rates
- Mean time-on-task (seconds)
- SUS — System Usability Scale
- UMUX-Lite score
- AI Trust Score (5-point survey)

QUALITATIVE METRICS

- Think-aloud observation notes
- Post-task satisfaction rating
- Open-ended interview themes
- Usefulness & trust probes

Study Materials & Protocol

HOW WE RAN THE STUDY

1

Introduction

Consent briefing
Study goals explained
Think-aloud method
No right/wrong answers

2

Task Completion

7 structured tasks
Live app (Streamlit)
Observer notes silently
Screen + audio recorded

3

Survey

SUS (10 questions)
UMUX-Lite (2 items)
AI Trust (custom 5-item)
Satisfaction ratings

4

Interview

Open-ended debrief
Most/least useful parts
Feature improvement ideas
Future usage intent

Quantitative Results

N = 5 · 7 TASKS · APRIL 2026

80%

Overall Task
Success Rate

75.5

SUS Score
Good ≥ 71

65.0

UMUX-Lite
Benchmark 70

76%

AI Trust
Score

Task-by-Task Results

Task	Success %	Assisted %	Failed %	Mean Time
T1 — Find most-reviewed city	100%	0%	0%	32s
T2 — Identify worst branch	80%	20%	0%	44s
T3 — Filter to single city	60%	20%	20%	94s
T4 — Navigate Co-Pilot	60%	40%	0%	97s
T5 — Generate AI insights	100%	0%	0%	93s
T6 — Download staff briefing	80%	20%	0%	24s
T7 — Adjust time range + regen	80%	20%	0%	110s
Overall	80%	17%	3%	71s

Qualitative Insights

THEMES FROM THINK-ALoud SESSIONS

✓ What Worked

Evidence-grounded AI outputs

3/5 users specifically praised action items that quoted actual reviews — increased trust significantly.

Dashboard clarity

All 5 participants understood the visualizations without any instruction — zero learning curve.

Ready-to-send briefing memo

4/5 said they would use the staff briefing in a real operational context.

✗ Key Frustrations

Row-click not discoverable (4/5)

Health table rows look read-only. No hover affordance. Most tried the dropdown instead.

Multiselect deselect-all confusion (3/5)

Users didn't realize they had to clear all cities before selecting one city.

AI spinner — no time expectation (3/5)

"It just said Analysing reviews... then nothing for 8 seconds. I thought it froze."

Interpretation & Discussion

WHAT THE RESULTS MEAN

✓ AI trust is earned through grounding, not just labeling

A SUS of 75.5 and AI trust score of 76% confirm users found the tool genuinely useful. The key differentiator was evidence citation — users who doubted AI summaries immediately changed their minds when action items quoted actual review text. This validates our core design choice to forbid hallucination at the prompt level.

⚠ Discoverability gaps masked by clean data presentation

While dashboard comprehension was flawless (100% T1 success), navigation tasks like row-click (T3/T4, 60% success) exposed that users don't explore — they follow visible affordances. Participants found 4 discoverability bugs in a 40-minute session, confirming even experienced analytics users need explicit UI signals.

📊 UMUX-Lite slightly below benchmark signals a gap between perceived usefulness and learnability

SUS=75.5 (Good) vs UMUX-Lite=65.0 (below 70) suggests users found the tool's outputs valuable but encountered friction learning the interaction model. This is consistent with the row-click and multiselect findings — the data quality shines, but the interaction vocabulary needs work.

Limitations, Risks & Ethics

CRITICAL CONSIDERATIONS

Small Sample Size (N=5)

Statistical power is low. Results indicate directional insights, not conclusive evidence. Findings need validation with a larger, more diverse group of real branch managers.

Ecological Validity Gap

Study tasks were artificial. Real managers face time pressure, interruptions, and partial data. Lab results likely overestimate performance in operational contexts.

Model Unpredictability & Prompt Sensitivity

Claude Haiku may respond differently to edge cases (single reviews, non-English text, spam). Outputs are not deterministic — identical inputs can produce varied summaries.

Privacy & Data Ethics

Only review text + star ratings are sent to Claude API — no PII, reviewer names, or location IDs. All LLM calls logged locally (SQLite) and never transmitted externally. Consent obtained from all participants.

J-Shaped Review Bias in Source Data

The Kaggle dataset over-represents 1-star and 5-star reviews. AI summaries may be skewed toward extreme sentiment. Aggregate star averages remain misleading without distribution context.

Conclusion & Future Work

WHAT WE BUILT · LEARNED · WHERE WE GO NEXT



MAIN TAKEAWAYS

1

Evidence-grounded AI beats black-box AI for trust: users rate outputs higher when they can verify the source evidence inline.

2

Human-AI collaboration works best as a decision support layer — not automation. All 5 users wanted final control; 4/5 said they'd adopt the tool in a real role.



Contribution to Human-AI Collaboration

SMCC demonstrates that grounding LLM outputs in retrieved evidence, not training-time generalization but is a practical, deployable approach to building AI tools workers actually trust and want to use in operational settings.



FUTURE IMPROVEMENTS



UX

Row hover tooltips
Editable briefing text
Progress bar with ETA



Data

Real-time Google Maps API
Sentiment trend over time
Competitor benchmarking



AI / Technical

Multi-branch comparison
GPT-4o fallback
Cost per-call dashboard



Deployment

Auth layer for managers
Role-based access control
Mobile-responsive layout

Acknowledgments & Contributions

FOR GRADING ONLY — NOT PRESENTED

Member	Primary Responsibilities
Pranav C	Backend architecture, Anthropic Claude API integration, LLM prompting, SQLite logging, data_loader.py
Jeet T	Streamlit UI layout, CSS theming (McDonald's brand), insight card rendering, download feature
Ashish G	Data engineering, Matplotlib/Seaborn charts, branch health table, trend calculation, user study coordination
All Members	User study design, SUS/UMUX-Lite surveys, qualitative analysis, CP4 report & evaluation

🔍 AI TOOLS & RESOURCES DISCLOSURE	
Tool	How Used
Anthropic Claude Haiku (claude-haiku-4-5-20251001)	Production AI model for insight generation within the SMCC app
Anthropic Claude (claude.ai / API)	Prompt engineering iteration, code debugging assistance, project report drafting
Streamlit	App framework — no AI involvement; standard Python library
Kaggle / kagglehub	Dataset source and download utility; no AI-generated data
GitHub Copilot	IDE code suggestions during development (accepted suggestions reviewed and tested by team)



Thank You!



I'm lovin' it 

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